

Build Solution

Given a use case that includes technical requirements, constraints or drivers, identify the considerations when designing and implementing API(s), both Salesforce as an API provider and Salesforce as an API consumer.

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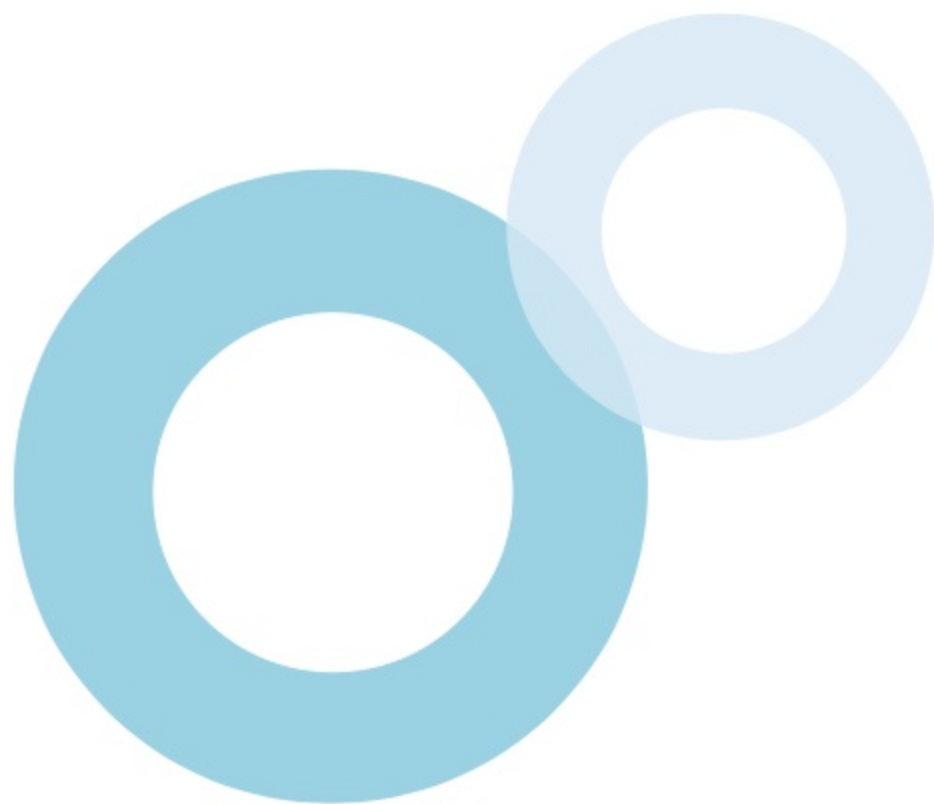
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After studying this topic, you should be able to:

- ✎ Identify the considerations for using Salesforce as an API provider.
- ✎ Identify the considerations for using Salesforce as an API consumer.
- ✎ Identify the considerations for designing and implementing an API for the given integration requirement.



Introduction

Salesforce offers various APIs for inbound integration, including **REST API, SOAP API, Bulk API 2.0, and Streaming API**. Various considerations and limitations apply to the use of an API. For example, Salesforce limits the number of API calls per 24-hour period, which depends on the edition and number of licenses.

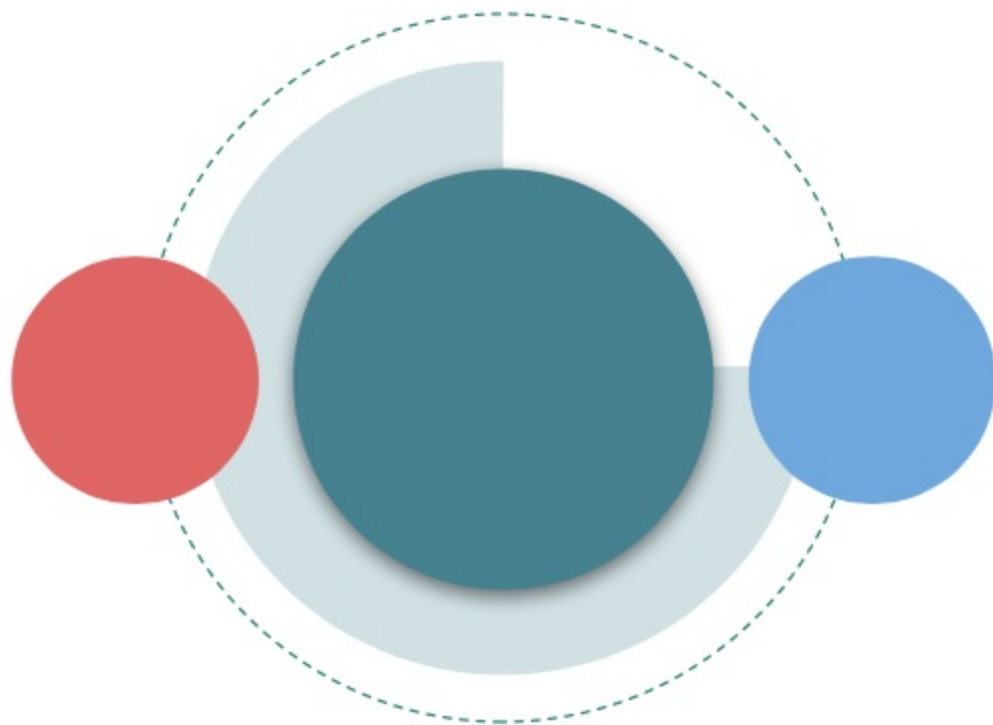
External Services or REST/SOAP callouts can be used to invoke an external API. Considerations related to **authentication, data format, and schema definition** are applicable to the use of an external service. Various **governor limits**, such as the maximum number of callouts allowed in an Apex transaction, apply to REST/SOAP callouts



Overview

Salesforce as API Provider

Salesforce offers various APIs for inbound integration, including **REST API, SOAP API, Bulk API 2.0, and Streaming API**. Various considerations and limitations apply to the use of an API.



Salesforce as API Consumer

Salesforce can consume an external API using **External Services or REST/SOAP callouts**. Considerations and governor limits that apply to them should be kept in mind.

Salesforce as API Provider

Salesforce offers various **APIs for inbound integration needs**. Various considerations apply, depending on the API selected for the integration.

❖ REST API

For invoking REST API from a remote system, considerations related to **resources, security, transaction/commit behavior, data volume, and governor limits** are applicable. A **session ID** obtained through a **valid login** is required for making REST API calls. **REST composite resources** can be used to perform multiple operations in a single call. The **'all or nothing'** behavior can be used to ensure that **a failure of one record change** causes **rollback of other record changes**. Up to **200 records** can be **created, updated or deleted** at a time. A limit is applicable to the number of **API calls per 24-hour period**, which depends on the **Salesforce edition** and **number of licenses**.



Salesforce as API Provider

Salesforce offers various **APIs for inbound integration needs**. Various considerations apply, depending on the API selected for the integration.

❖ SOAP API

For invoking SOAP API from a remote system, considerations related to **generated WSDL, security, transaction/commit behavior, data volume, and governor limits** are applicable. A **session ID** obtained through a **valid login** is required for making SOAP API calls. Two WSDLs, namely **Enterprise WSDL** and **Partner WSDL**, are available for remote systems. Enterprise WSDL is a **strongly typed WSDL** that is **specific** to a Salesforce org. On the other hand, Partner WSDL is **loosely typed** and **not specific** to a Salesforce org. The **'all or nothing'** behavior is available to roll back all records if any error occurs. Up to **200 records** can be **created, updated or deleted** at a time. A limit is applicable to the number of **API calls per 24-hour period**, which depends on the **Salesforce edition** and **number of licenses**.



Salesforce as API Provider

Salesforce offers various **APIs for inbound integration needs**. Various considerations apply, depending on the API selected for the integration.

❖ BULK API 2.0

For invoking Bulk API 2.0 from a remote system or client, considerations related to **data volume, security, and governor limits** are applicable. A **session ID** obtained through a **valid login** is required for making Bulk API 2.0 calls. If a data operation requires processing **more than 2,000 records**, it is a good candidate for the use of Bulk API 2.0. It can be used to **query, insert, update, upsert, or delete** hundreds of thousands to millions of records asynchronously by submitting a number of batches. Bulk API 2.0 is **REST-based**, so the same calling mechanisms apply. Up to **15,000 batches** can be submitted per 24-hour rolling period. Each batch can contain a maximum of **10,000 records**. And up to **150 million records** can be uploaded per 24-hour rolling period.



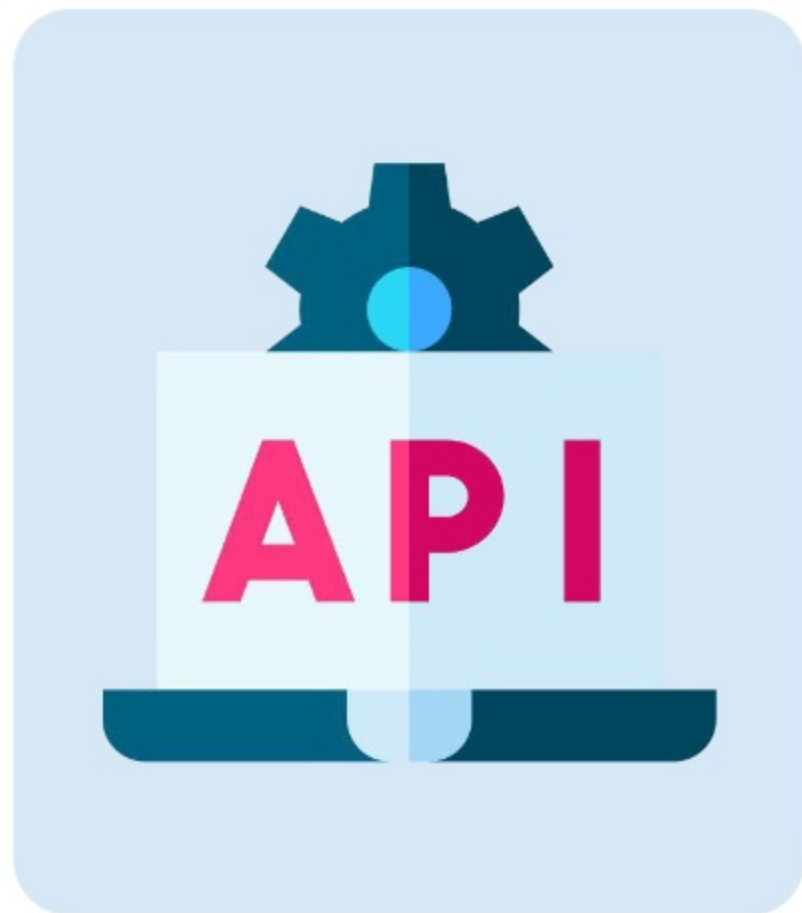
Salesforce as API Provider

Salesforce offers various **APIs for inbound integration needs**. Various considerations apply, depending on the API selected for the integration.

❖ STREAMING API

Considerations related to **notifications, security, and governor limits** are applicable to the use of Streaming API for streaming notifications to the Salesforce user interface. The **delivery and order of notifications** are **not guaranteed**. **Record changes made by Bulk API do not generate notifications**. Salesforce recommends the use of **HTTPS** to connect to Streaming API. **Standard-volume platform events** allow up to **100,000 events** to be published per hour. As for **high-volume platform events**, up to **250,000 events** can be published per hour. The **maximum event message size** is **1 MB**.

Note: **standard-volume** platform events, which are already considered legacy custom events, are scheduled for **retirement** on the **Summer '27** release.



Salesforce as API Consumer

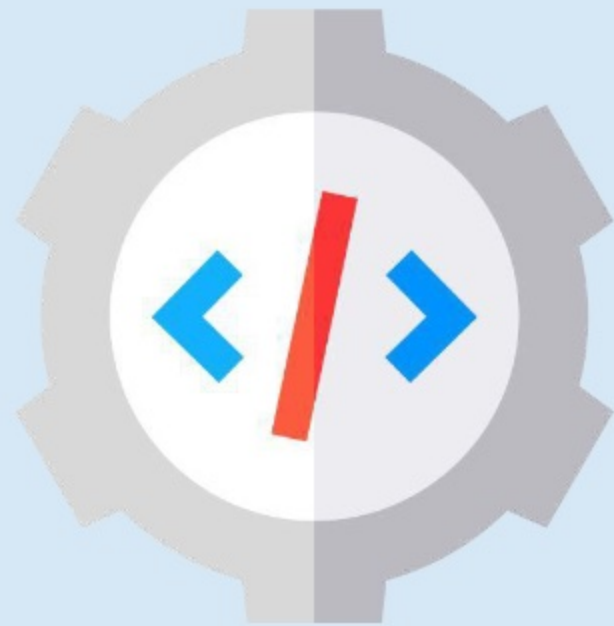
Certain considerations are applicable when Salesforce consumes an external API using **External Services** or a **REST/SOAP callout**.

❖ EXTERNAL SERVICES

There are certain semantic, service, and usage constraints for integrating an external service using External Services. External Services uses **Named Credentials** for **authentication**. The **supported media types** for request and response content serialization are **application/json**, **application/x-www-form-urlencoded**, and **text/plain**. The **supported schema formats** are:

- a) **OpenAPI 2.0, JSON schema format**
- b) **OpenAPI 3.0, JSON schema format**
- c) **UTF-8**, with full character set for names and identifiers

Various restrictions related to **schema definition** are also applicable. For example, a schema can have up to **10,000,000 characters (10 MB)**, and there can be up to **150 External Services registrations** in an org.

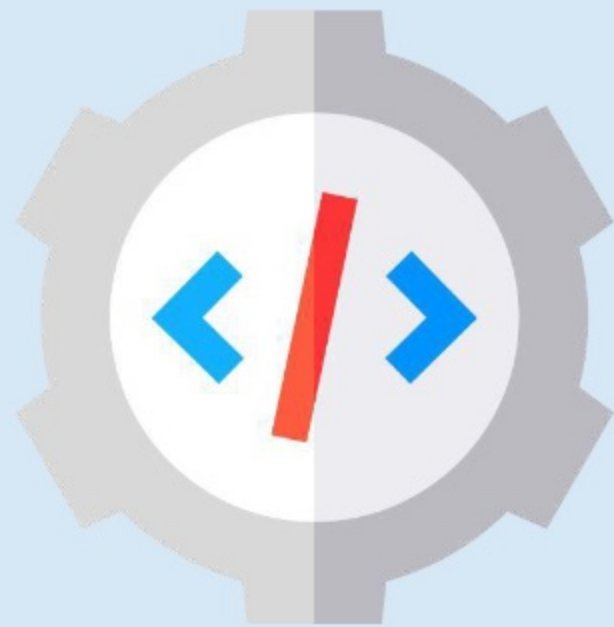


Salesforce as API Consumer

Certain considerations are applicable when Salesforce consumes an external API using **External Services** or a **REST/SOAP callout**.

❖ REST/SOAP CALLOUTS

Various limits and limitations apply when executing a REST or SOAP callout from Apex code. A single Apex transaction can contain a maximum of **100 HTTP requests or web service calls**. Although the **default timeout** of a callout is **10 seconds**, a **custom timeout** of up to **120,000 milliseconds** can be defined. However, the **maximum cumulative timeout** for all the callouts in a transaction is **120 seconds**. There are also other governor limits, such as the limit of **5 seconds** on **long-running requests**. In addition, a callout cannot be made if there are other **pending operations** in a transaction, including **DML statements**, **asynchronous Apex**, **scheduled Apex**, or **sending email**. When Salesforce is in **read-only mode** during a maintenance activity, it is recommended to prevent a callout to avoid inconsistent program behavior.

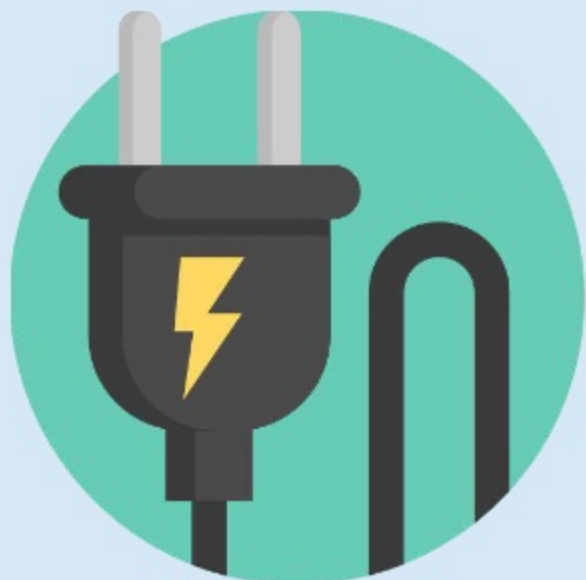


Mulesoft

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Integration Summary

Integration is the process of combining various **separate components or subsystems** into a **cohesive and functioning system** that can effectively communicate and work together.



❖ INTEGRATION COMPONENTS

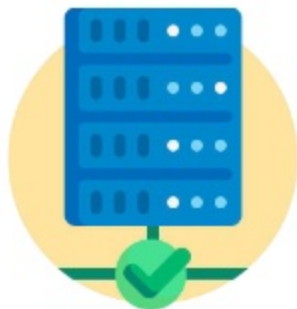
Integration is composed of two components. First is **physical integration**, which is the actual linking of the systems and applications to support communication. Second is the **business integration**, which is identifying the data that needs to be integrated, its frequency, and how it aligns with existing business processes.

❖ INTEGRATION OPTIONS

There are several options for integrating Salesforce with other systems such as using **pre-built connectors**, building a **custom solution** using standard Salesforce APIs (SOAP, REST, Streaming API, etc.), using a **middleware tool** such as Mulesoft, or resorting to **other options**.

Mulesoft

Mulesoft is a tool that provides resources for **system integration**. It promotes **API-led connectivity**, which is a methodical way to connect applications that organizes APIs into **three broad categories**:



SYSTEM APIS

System APIs are used to communicate directly with **backend systems** such as databases, web services, SaaS/On-premise apps, social apps, etc.



PROCESS APIS

Process APIs interact with and process data collected from a **single system or multiple systems** and break down data fragmentation by unifying data.

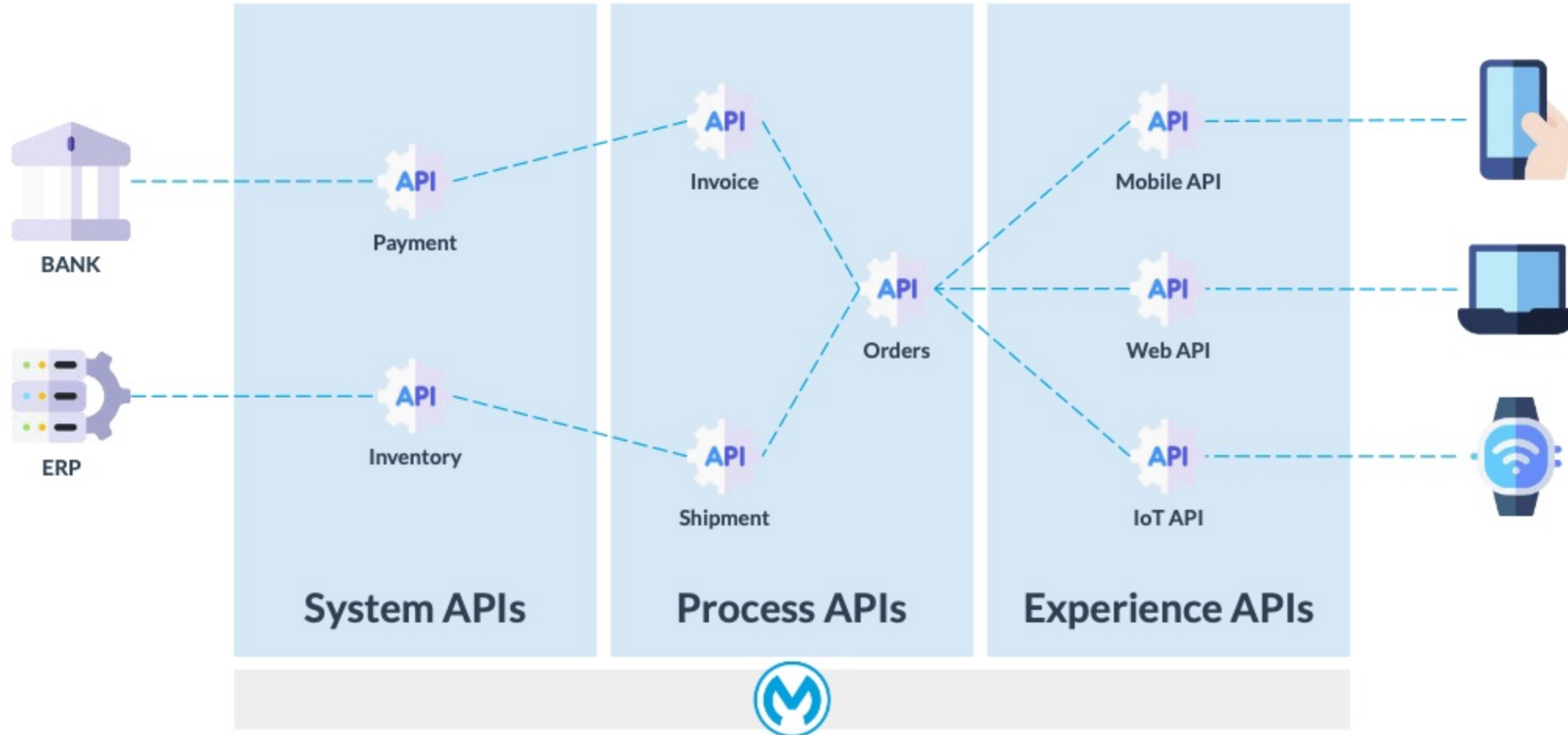


EXPERIENCE APIS

Experience APIs handle and ready data for **consumption of end users** through a medium or service such as a desktop browser, mobile app, or another platform.

Mulesoft

The following shows an example **API-led integration pattern** using Mulesoft.



Mulesoft Anypoint Platform

The Mulesoft Anypoint Platform provides tools for supporting the **API development lifecycle** and is comprised of the following main **components**.



❖ ANYPOINT DESIGN CENTER

Mulesoft's Anypoint **Design Center** can be used to **create** an API specification. To verify that the API works correctly, it can be **tested** using the **Mocking Service**. When ready, the API can be **published** in the **Anypoint Exchange**.

❖ ANYPOINT MANAGEMENT CENTER

Mulesoft's Anypoint Management Center is a central location for **managing integrations, applications**, and **all aspects of an API** such as initial deployment, ongoing operations, and maintenance.

Mulesoft Anypoint Platform

The Mulesoft Anypoint Platform provides tools for supporting the **API development lifecycle** and is comprised of the following main **components**.



❖ ANYPOINT EXCHANGE

The Anypoint Exchange is a marketplace or central repository for **storing private assets** and **accessing publicly shared resources** such as APIs, connectors, templates, and other integration assets.

❖ MULE

Mule is the **runtime engine** of the Anypoint Platform and is a lightweight Java-based **enterprise service bus** (ESB) and **integration platform**.

❖ RUNTIME SERVICES

Runtime services is a **suite of platform services** providing security, reliability, scalability, and high-availability (e.g. data storage, message queuing, etc.).

Mulesoft Composer

MuleSoft Composer for Salesforce, which is also known as **Flow Integration**, is part of the Salesforce Flow suite of drag-and-drop tools that can be used for **integration automation**.



❖ INTEGRATION TOOL

The Composer facilitates in building **process automation for data**, which may be stored across various systems, using clicks instead of code in the org.

❖ WORKFLOW

For example, when **data changes** are made in an external system (Google Sheets, Tableau, etc.), Composer can be configured to **detect the change** and **perform a specific action** such as relaying the change to the org.

❖ USE CASES

Example use cases for Composer is quote-to-cash processes (**ERP integration**), case escalation (**case management systems**), or Salesforce to Salesforce integration (**data synchronization**).

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Printers is a company that manufactures and sells various types of inkjet and laser printers to B2B and B2C customers in the United States. It recently started using Unlimited edition of Salesforce for managing products and employee information. Other enterprise systems used by the company include an ERP system and a KMS (Knowledge Management System). The ERP system stores data about customers, including accounts and contacts. The KMS contains thousands of articles about the printers sold by the company.

The company wants to integrate the systems with one another to make it easier for internal users to access or process information in other systems. Various technical requirements have been defined. A team of developers is preparing the solution design document that describes all aspects of the required integration solutions. A solution architect has been appointed, whose main responsibility is to identify the considerations for designing and implementing the APIs that would meet the technical requirements.

Requirements & Solutions

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Requirement 1



SCENARIO

The ERP system is used by the company's sales reps to manage information about customers and products purchased by them. They often receive calls from customers who would like to make a bulk purchase directly from the company. When a sales rep needs to add a product to a customer's account in the ERP system, they need to log in to Salesforce to get the detailed description of the product. The sales director would like to simplify the process of obtaining the product information by integrating the two systems. The use of REST API is being considered by the IT team to allow ERP users to retrieve the required information from Salesforce. The solution architect needs to identify any related considerations.



Solution 1



SOLUTION

When a sales rep requires product information stored in Salesforce, a REST API call to Salesforce can be utilized. A valid **session ID** must be obtained using **valid credentials** for making the REST API call. The **Query** resource can be used to execute a SOQL query that returns the information. However, it would be important to consider the **number of REST API calls** required **per 24-hour period**. Since the company uses **Unlimited edition** of Salesforce, the total number of API calls that can be made per 24-hour period can be no more than **100,000 + (number of licenses x calls per license type) + purchased API Call Add-Ons**. Salesforce APIs that count toward this limit include **REST API, SOAP API, Bulk API, and Bulk API 2.0**, so the number of calls made using any of these APIs should be considered.

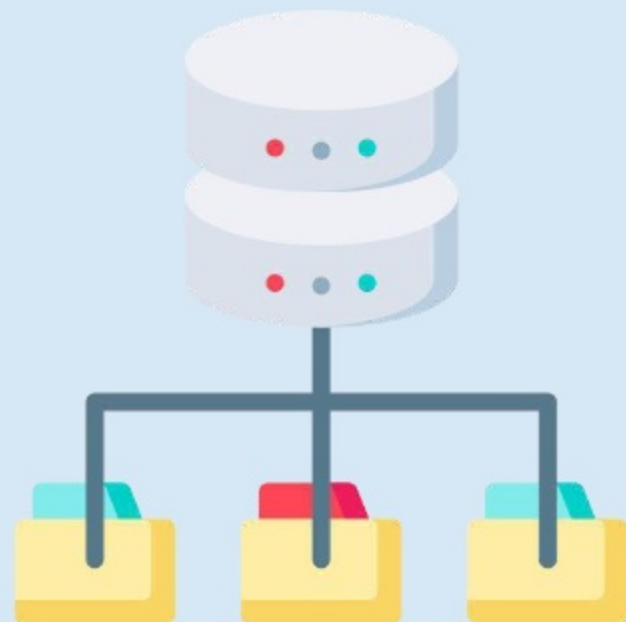


Requirement 2



SCENARIO

A developer has been assigned to build an external application that would allow specific users to extract data from the ERP system and load it into Salesforce. The application will need to support loading hundreds of thousands of records, such as accounts and contacts. The solution architect needs to describe any relevant considerations for using Bulk API 2.0 to meet this requirement.

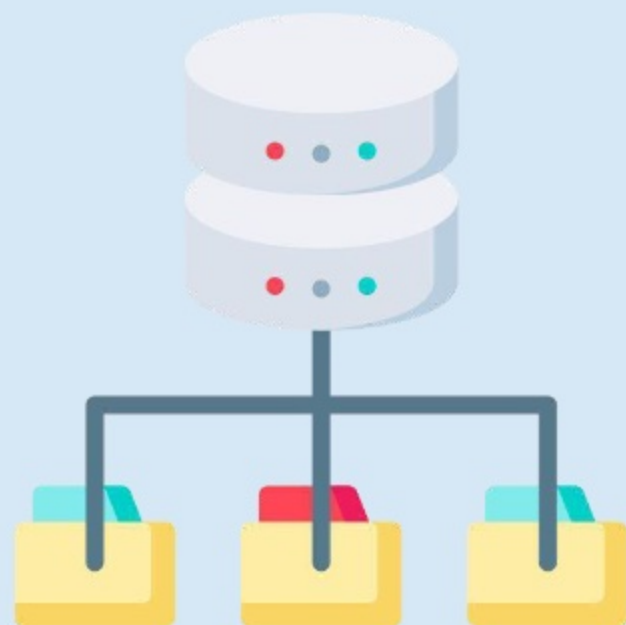


Solution 2



SOLUTION

Since Bulk API 2.0 is REST-based, the same calling mechanisms would apply. But since Bulk API 2.0 is asynchronous, the request is processed in the background. A user of the application must come back for the results later. A valid **session ID** must be obtained using **valid credentials** for making a Bulk API 2.0 call. A bulk job needs to be created, data must be added to the job, and then the job can be closed to let Salesforce process the records. Up to **15,000 batches** of records can be submitted per 24-hour rolling period. Each batch can contain a maximum of **10,000 records**. And up to **150 million records** can be uploaded per 24-hour rolling period.

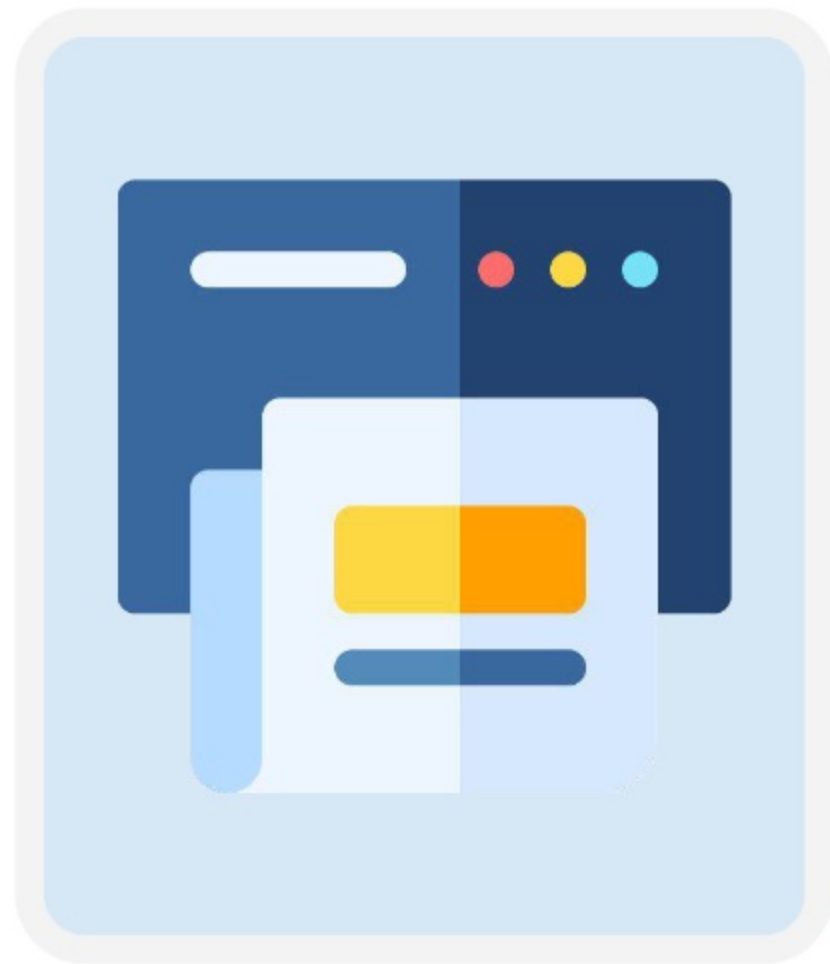


Requirement 3



SCENARIO

The support department of the company has recently started using Salesforce for case management. While working on a case, a support agent should be able to retrieve a related article stored in the Knowledge Management System (KMS) and attach it to the case. The KMS has a SOAP web service that can support the integration for this requirement. The solution design consists of a quick action that invokes a Lightning component for a SOAP callout to the KMS. The solution architect must identify any considerations that should be kept in mind while implementing this solution.



Solution 3



SOLUTION

To invoke a SOAP callout from Salesforce, it would be necessary to obtain the **WSDL file** of the web service from the KMS.

WSDL2Apex on the **Apex Classes** page in Salesforce Setup should be used to generate Apex classes from the WSDL document.

These include **stub and type classes** that will allow calling the web service from Apex code. Also, the SOAP callout must be performed **before any DML operations**, such as creating an article in Salesforce based on the data received from the KMS. This is because Salesforce does not allow a callout if there are other **pending operations** in a transaction, such as a **DML statement**.

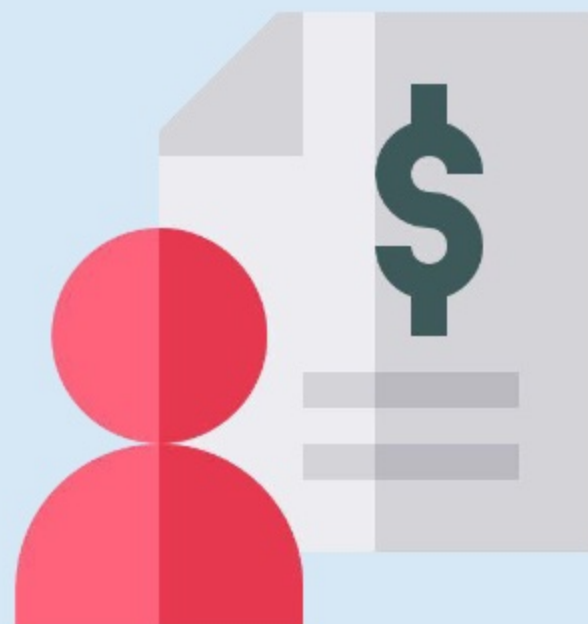


Requirement 4



SCENARIO

The company uses a payroll application that contains timesheet and pay data of employees. Currently, employees can access basic information about their company role on their record in Salesforce. A custom object named 'Employee_c' has been created. The HR director would like to give them access to the information in the payroll application which has a REST-based API. An external service invoked by a flow is being considered for this use case. Due to the sensitivity of the data stored in the payroll application, the solution architect needs to provide any considerations related to authentication and data format.



Solution 4



SOLUTION

The external service must use a **Named Credential** for **authentication**. It **does not support** authentication data stored in a flow. The **URL of the callout endpoint** and its required **authentication parameters** can be defined in the named credential. The data retrieved from the payroll application can be in **JSON format**, which can be parsed, encrypted, stored if required, and made accessible only to the users who should be able to view it. Any applicable schema limitations must also be considered. For example, the schema definition for the registered external service can have up to **10,000,000 characters (10.0 MB)**.



Requirement 5



SCENARIO

An Experience Cloud site has been built for the company's sales users. It needs to be integrated with the ERP system to allow certain sales users to update account information stored in the ERP system. Users should be able to access a site page, retrieve information about a particular account, and perform an update after entering details in fields on the page. The use of a REST callout from a Lightning web component is being considered for this use case. The solution needs to identify any related considerations.



Solution 5



SOLUTION

There are two important considerations related to the use of REST callouts for this use case. The **default timeout** of a callout from Apex code is **10 seconds**, but a **custom timeout** of up to **120,000 milliseconds** can be defined. Furthermore, if any DML or other Salesforce operations need to be performed in Salesforce, it is important to consider that a callout cannot be made if there are other **pending operations** in an Apex transaction, including **DML statements**, **asynchronous Apex**, **scheduled Apex**, and **sending email**.



Learn More

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-  [Callout Limits and Limitations](#)
-  [Bulk API and Bulk API 2.0 Limits and Allocations](#)
-  [External Services Considerations](#)
-  [Get Started with Anypoint Platform](#)
-  [Build Great APIs and Integrations with MuleSoft](#)